

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 23

User Plane

صفحه کاربر

GTP-U · N3 · N9 · N6



Lecture Notes

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1 User Plane Concept

The **User Plane** (also called the **Data Plane** or **U-Plane**) is responsible for carrying actual user data through the mobile network. This includes:

- **IP packets** (web browsing, app data, file transfers)
- **Voice data** (VoLTE/VoNR encoded speech frames)
- **Video streams** (real-time video, streaming content)
- **Any application-layer payload**

The user plane is distinct from the control plane — while the control plane sets up, manages, and tears down connections, the user plane is the “pipe” through which data actually flows once those connections are established.

Key characteristics:

- High throughput requirement
- Low latency sensitivity
- Stateless forwarding (once tunnels are established)
- QoS-aware packet handling
- Transparent to signaling — carries data without interpreting it

2 GTP-U Protocol (GPRS Tunneling Protocol – User Plane)

GTP-U (3GPP TS 29.281) is the core tunneling protocol used to transport user data between network nodes across all generations (3G, 4G, 5G).

2.1 Header Structure

The GTP-U header has an **8-byte mandatory portion** plus optional extension headers:

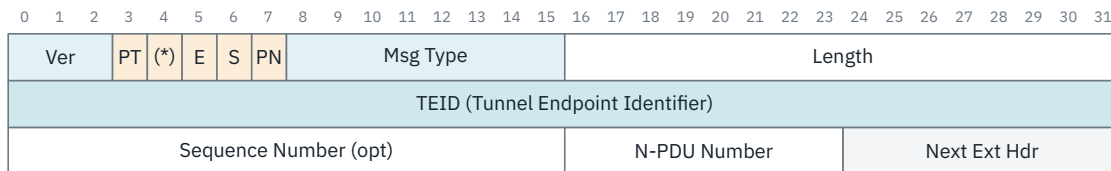


Figure 23.1 GTP-U header

Field	Bits	Description
Version	3	Always 001 (GTPv1)
PT (Protocol Type)	1	1 = GTP, 0 = GTP' (charging)
Reserved (*)	1	Must be 0
E (Extension Header)	1	Extension header follows
S (Sequence Number)	1	Sequence number field present
PN (N-PDU Number)	1	N-PDU number field present
Message Type	8	0xFF = T-PDU (user data), others = signaling
Length	16	Payload length (excludes mandatory 8 bytes)
TEID	32	Tunnel Endpoint Identifier

2.2 TEID (Tunnel Endpoint Identifier)

The **TEID** is a 32-bit value that uniquely identifies a GTP tunnel endpoint at a receiving node:

- Each direction of a tunnel has its own TEID
- The **receiving node** assigns the TEID (tells the sender “use this TEID when sending to me”)
- TEID = 0 is reserved for control plane initial messages
- A node may have thousands of TEIDs active simultaneously (one per bearer/session per direction)

Example:

```
gNB tells UPF: "For downlink data to UE-X, use TEID = 0x00A1B2C3 on N3"
UPF tells gNB: "For uplink data from UE-X, use TEID = 0x00D4E5F6 on N3"
```

2.3 Encapsulation / Decapsulation

At each GTP-U hop, the user's IP packet is **encapsulated** inside:

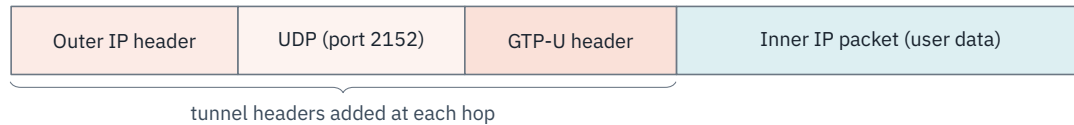


Figure 23.2 GTP-U encapsulation at each hop

- **Encapsulation** (sending node): Prepends GTP-U header + UDP + outer IP
- **Decapsulation** (receiving node): Strips outer IP + UDP + GTP-U header, forwards inner IP
- At intermediate nodes: may decapsulate and re-encapsulate with new TEID

2.4 Sequence Numbers and Extension Headers

Sequence Numbers (16-bit):

- Optional but used for reordering at tunnel endpoints
- Critical during handover (ensures no packet loss/reorder)
- Incremented per PDU per tunnel

Extension Headers:

- Chained via “Next Extension Header Type” field
- Each extension: `[Length (1 byte)][Content (variable)][Next Ext Hdr Type (1 byte)]`

Common extension headers:

Type	Name	Purpose
0x85	PDU Session Container	5G QoS Flow Identifier (QFI), reflective QoS
0xC0	PDCCP PDU Number	For lossless handover
0x40	UDP Port	Additional multiplexing

3 User Plane in LTE (4G)

3.1 End-to-End Path

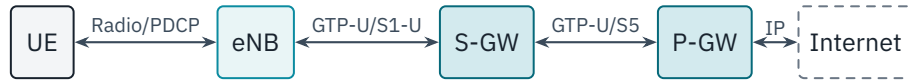
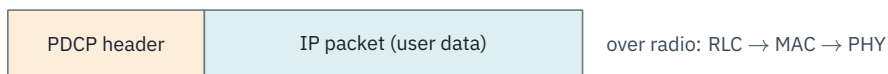


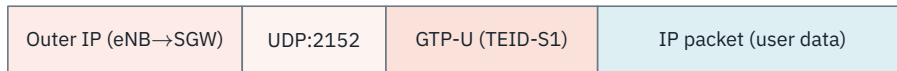
Figure 23.3 End-to-end LTE user-plane path

3.2 Packet Headers at Each Segment

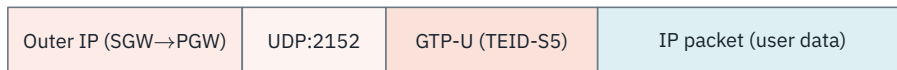
Segment 1: UE → eNB (radio interface / Uu)



Segment 2: eNB → S-GW (S1-U)



Segment 3: S-GW → P-GW (S5/S8)



Segment 4: P-GW → Internet

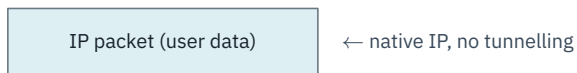


Figure 23.4 Packet headers on each LTE user-plane segment

3.3 LTE Bearer Model

- **Default Bearer:** Always-on, established at attach (one per PDN connection)
- **Dedicated Bearer:** Additional QoS, established on demand (e.g., VoLTE)
- Each bearer = unique set of TEIDs on S1-U and S5

4 User Plane in 5G

4.1 End-to-End Path

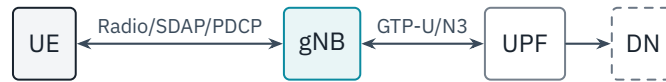


Figure 23.5 End-to-end 5G user-plane path

4.2 5G Reference Points for User Plane

Interface	Between	Transport	Purpose
N3	gNB ↔ UPF	GTP-U over UDP/IP	RAN to Core user data
N9	UPF ↔ UPF	GTP-U over UDP/IP	Inter-UPF forwarding
N6	UPF ↔ DN	Native IP	Exit to data network

4.3 SDAP Layer (Service Data Adaptation Protocol)

The **SDAP** layer is new in 5G NR (not present in LTE) and sits above PDCP:

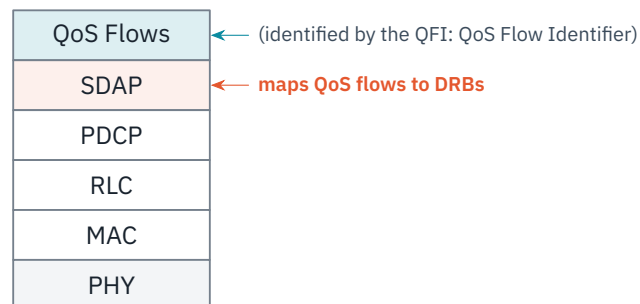


Figure 23.6 NR user-plane stack with the new SDAP layer

SDAP Functions:

- **Downlink:** QoS Flow → DRB mapping (based on QFI in GTP-U extension header)
- **Uplink:** DRB → QoS Flow marking (stamps QFI on uplink packets)
- **Reflective QoS:** UE mirrors the DL mapping for UL without explicit signaling

4.4 Multiple UPFs Architecture

5G supports **UPF chaining** for flexible traffic routing:

UPF Type	Role	Use Case
I-UPF (Intermediate)	Transit/branching point	Handover anchor, traffic steering
PSA-UPF (PDU Session Anchor)	Session anchor, IP allocation	Stable IP address point

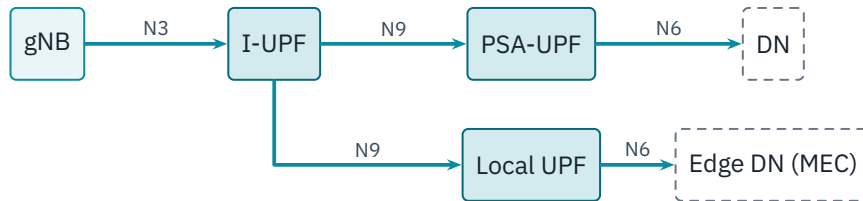


Figure 23.7 UPF chaining with a local breakout to the edge

5 UPF Functions (3GPP TS 23.501)

The UPF is controlled by the SMF via the **N4 interface** using the **PCFP protocol** (Packet Forwarding Control Protocol, TS 29.244).

5.1 PCFP Rules

Rule	Acronym	Function
Packet Detection Rule	PDR	Match incoming packets (by TEID, IP, port, QFI, etc.)
Forwarding Action Rule	FAR	What to do (forward, drop, buffer, duplicate)
QoS Enforcement Rule	QER	Rate limiting (MBR/GBR), gating, marking
Usage Reporting Rule	URR	Volume/time measurement, triggers for reporting
Buffering Action Rule	BAR	Buffer DL packets when UE is idle, notify SMF

5.2 How Rules Work Together

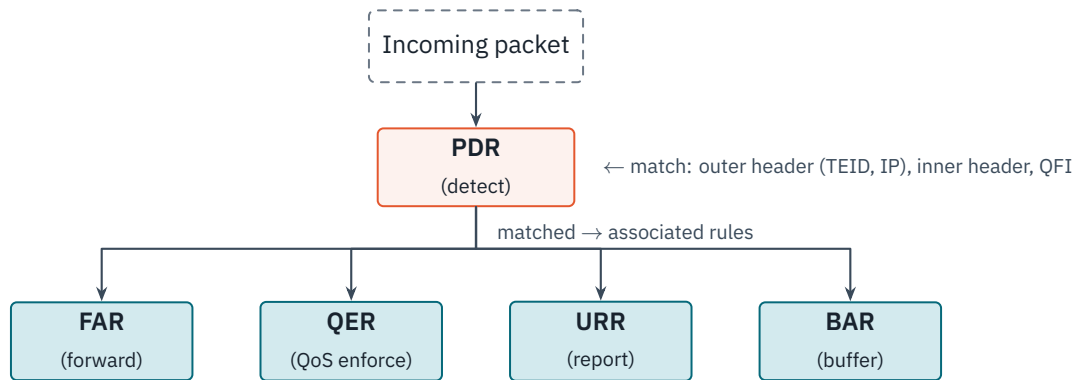


Figure 23.8 How the PFCP rules work together in the UPF

5.3 N4/PCFP Interface

- **SMF** → **UPF**: Session Establishment/Modification/Deletion Request
- **UPF** → **SMF**: Session Report (usage reports, DL data notification)
- PFCP uses **SEID** (Session Endpoint Identifier) — analogous to TEID but for control
- Runs over UDP port 8805

5.4 Complete UPF Function List

1. Packet routing & forwarding
2. Packet inspection (DPI for application detection)
3. UL/DL rate enforcement (per QoS flow)
4. Traffic usage reporting (for charging)
5. DL packet buffering & notification
6. QoS handling (DSCP marking, policing)
7. Lawful intercept (UP collection)
8. Transport level packet marking (outer header DSCP)

6 Packet Trace Example: Downlink Data

Scenario: Server on Internet sends data to UE

6.1 Step-by-Step

Step 1: Internet → UPF (N6)

IP header
Src: 93.184.216.34 (server)
Dst: 10.0.0.1 (UE IP)
TCP/UDP header
Application data (payload)

Step 2: UPF → gNB (N3, GTP-U)

Outer IP header
Src: 192.168.100.1 (UPF N3 IP)
Dst: 192.168.200.1 (gNB N3 IP)
UDP header (Src: ephemeral, Dst: 2152)
GTP-U header
TEID: 0x00A1B2C3 (assigned by the gNB)
Ext Hdr: PDU Session Container
QFI: 9 (QoS Flow ID)
Inner IP header
Src: 93.184.216.34 (server)
Dst: 10.0.0.1 (UE IP)
TCP/UDP + application data

Step 3: gNB → UE (radio)

SDAP header (QFI: 9)
PDCP header (SN, integrity info)
IP header
Src: 93.184.216.34
Dst: 10.0.0.1
TCP/UDP + application data

RLC segmentation → MAC mux
→ PHY (radio transmission)

Step 4: at the UE

IP header
Src: 93.184.216.34
Dst: 10.0.0.1
TCP/UDP + application data

delivered to the application layer

Figure 23.9 A downlink packet at each step from the Internet to the UE

6.2 Encapsulation Layer Diagram

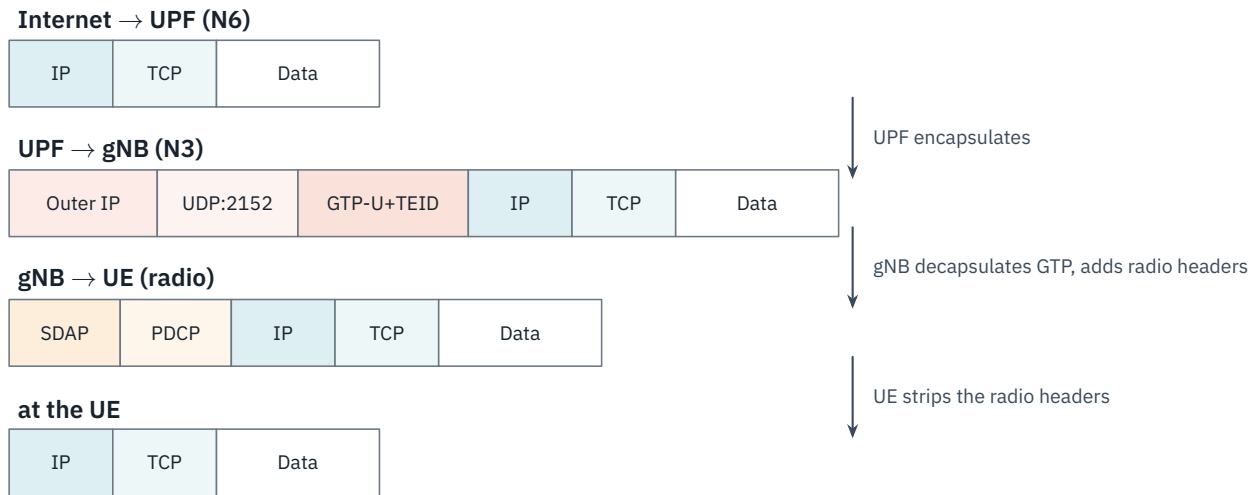


Figure 23.10 Headers of a downlink packet at each hop

7 User Plane Optimization

7.1 Dual Connectivity (DC) – Split Bearer

In **EN-DC** (E-UTRA–NR Dual Connectivity) or **NR-DC**:

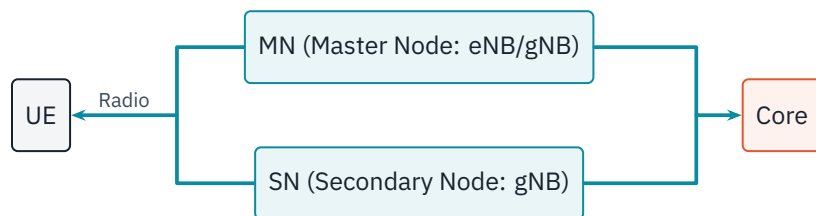


Figure 23.11 Dual connectivity with a split bearer

Split Bearer types:

Type	Description	GTP-U path
MCG Bearer	Data only via Master Node	UPF ↔ MN
SCG Bearer	Data only via Secondary Node	UPF ↔ SN
Split Bearer	Data split across both nodes	UPF ↔ MN ↔ SN (MN routes)

- Increases throughput by aggregating radio resources from two nodes
- PDCP layer at MN handles reordering for split bearers

7.2 Multi-Path UPF

Multiple UPFs in the data path for:

- **Load balancing:** Distribute traffic across UPFs
- **Redundancy:** N9 links between UPFs for resiliency
- **Geographic optimization:** Route traffic through nearest UPF

7.3 Edge Computing (Local Breakout)

5G enables **traffic steering** for MEC (Multi-access Edge Computing):

UL Classifier (UL CL):

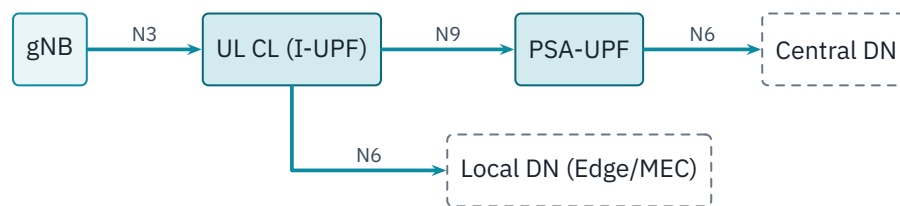


Figure 23.12 Uplink classifier for edge breakout

- UL CL inspects UL packets and routes matching traffic to local DN
- Non-matching traffic continues to central DN
- **Same UE IP address** for both paths (transparent to UE)

Branching Point:

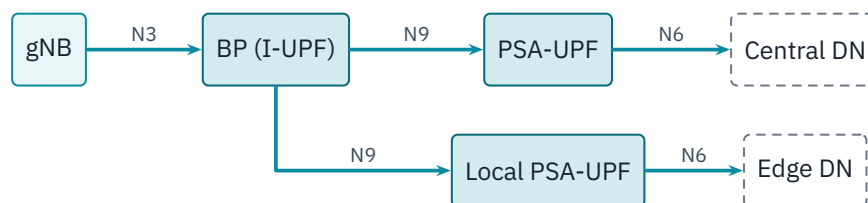


Figure 23.13 Branching point for IPv6 multi-homing

- Different PDU session anchors for local vs. remote
- May use different UE IP addresses

Benefits of Edge Computing:

- Ultra-low latency (< 10 ms RTT)
- Reduced backhaul bandwidth
- Local data processing (privacy, compliance)
- Use cases: AR/VR, autonomous driving, industrial IoT

Complete 5G User Plane Path Diagram

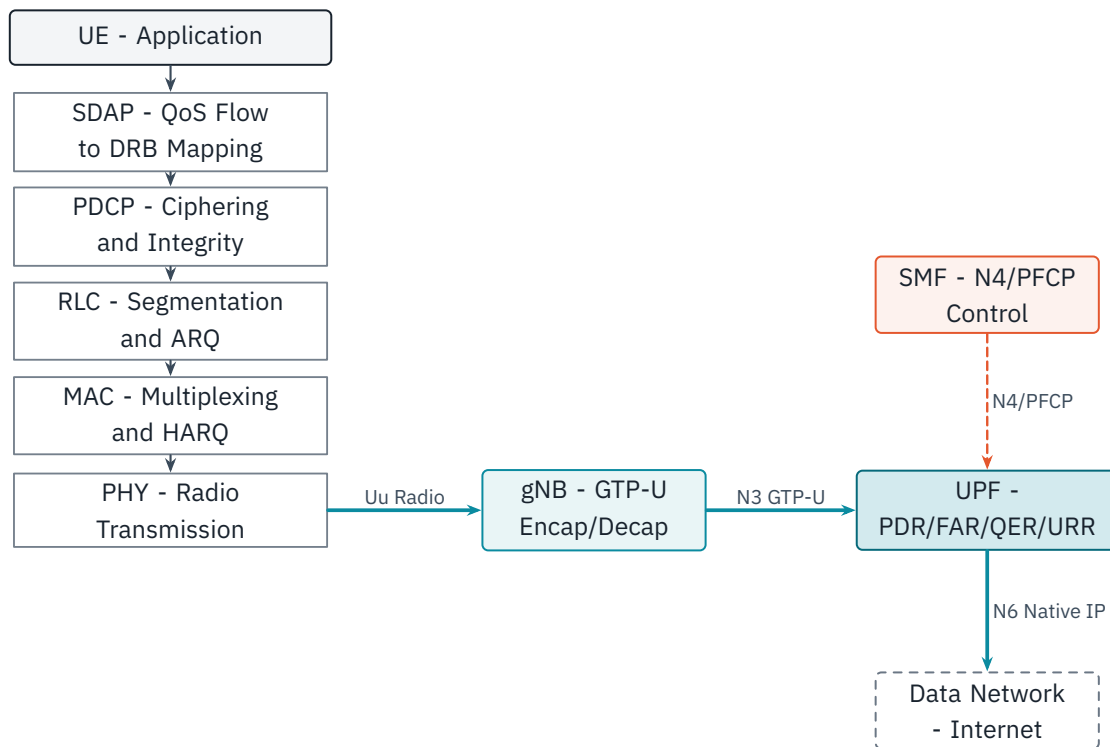


Figure 23.14 Complete 5G user-plane path

Summary Table: User Plane Across Generations

Aspect	3G (UMTS)	4G (LTE)	5G (NR)
Tunnel Protocol	GTP-U	GTP-U	GTP-U
RAN→Core Interface	Iu-PS	S1-U	N3
Core Internal	Gn (SGSN↔GGSN)	S5/S8 (SGW↔PGW)	N9 (UPF↔UPF)
Core→DN	Gi	SGi	N6
QoS Mechanism	Traffic classes	Bearers (QCI)	QoS Flows (QFI)
Radio QoS Layer	—	—	SDAP
Gateway Function	GGSN	P-GW	UPF
Control of UP	—	—	PCF (N4)
Edge Support	No	Limited	Native (UL CL, BP)

Key Takeaways

1. **GTP-U** is the universal user plane tunnel — same protocol from 3G to 5G
2. **TEID** identifies each tunnel endpoint — assigned by the receiver
3. **5G separates control and user plane** (CUPS) — SMF controls UPF via PCF/N4
4. **SDAP** is the new 5G layer that maps QoS flows to radio bearers
5. **UPF rules (PDR/FAR/QER/URR/BAR)** provide programmable packet processing
6. **Edge computing** is natively supported via UL Classifier and Branching Point
7. User plane optimization (DC, multi-UPF, edge) enables diverse 5G use cases